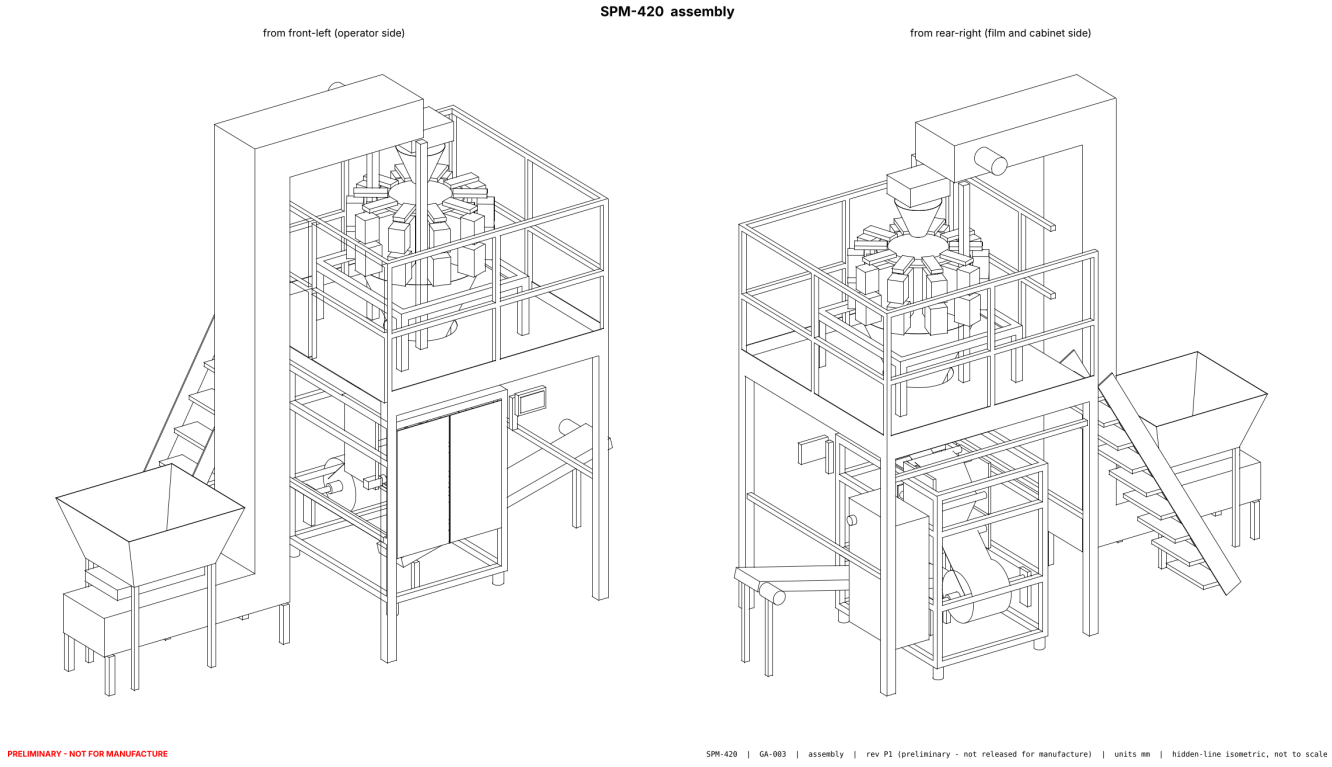
NULL / TECH ENGINEERING STUDIO

PROJECT OVERVIEW AND SCOPE OF SUPPLY

SPM-420 Automatic Snack Filling and Packaging Line

Multihead weigher with a vertical form-fill-seal bagger for potato chips, crispy snacks, crackers and small biscuits



Interactive 3D model and simulation: spm-420.vercel.app

PREPARED BY	DOCUMENT	REVISION	DATE	UNITS
NULL-TECH Engineering Studio	Project overview	P1, preliminary	11 November 2026	mm, kg, s, USD / IQD

Status: design stage

This package is a preliminary design, checked by engineering calculation and computer simulation. No part of it has been built, run on a real controller or measured. All performance figures are calculated or simulated and are marked as such. Drawings are not released for manufacture.

CONTENTS

1. Summary
2. How the machine works, with the 3D simulation
3. Layout and dimensions
4. What the design package contains
5. What has been verified, and how
6. What the project must supply
7. Items that can still be added to the package
8. Estimated budget
9. Open issues and assumptions
10. Recommended next steps

1. Summary

The SPM-420 is a complete snack packaging line. Product is lifted to a 10-head combination weigher, which drops exact charges into a vertical bagger that forms pillow bags from roll film, fills, seals and cuts them. The bagger is designed in full; the weigher and the elevator are specified as purchased units with defined interfaces.

The reference for the design was a short manufacturer video of a chips and biscuits filling machine. The video frames could not be viewed while the design was prepared, so the reference analysis is based on the video's title, the publisher's catalogue and published data for this machine class. The design follows the most common configuration for snacks, a multihead weigher feeding a vertical form-fill-seal bagger. Please confirm this against the video before detail design begins.

4450 × 1900 × 3900 overall line, mm (L × W × H)	**6050 × 4100 mm** floor area with access zones (24.8 m²); 4400 mm clear height	**40 bags/min** rated, 40 g chips in a 160 × 230 mm bag; 49 calculated maximum
80–200 × 80–300 mm bag width × length; film web up to 420 mm	**4.7 kW, 0.19 m³/min** connected load at 400/230 V; compressed air at 0.6 MPa	**≈ 56,000 USD** semi-industrial estimate, ±30 %; ≈ 73.8 million IQD

Maturity at a glance

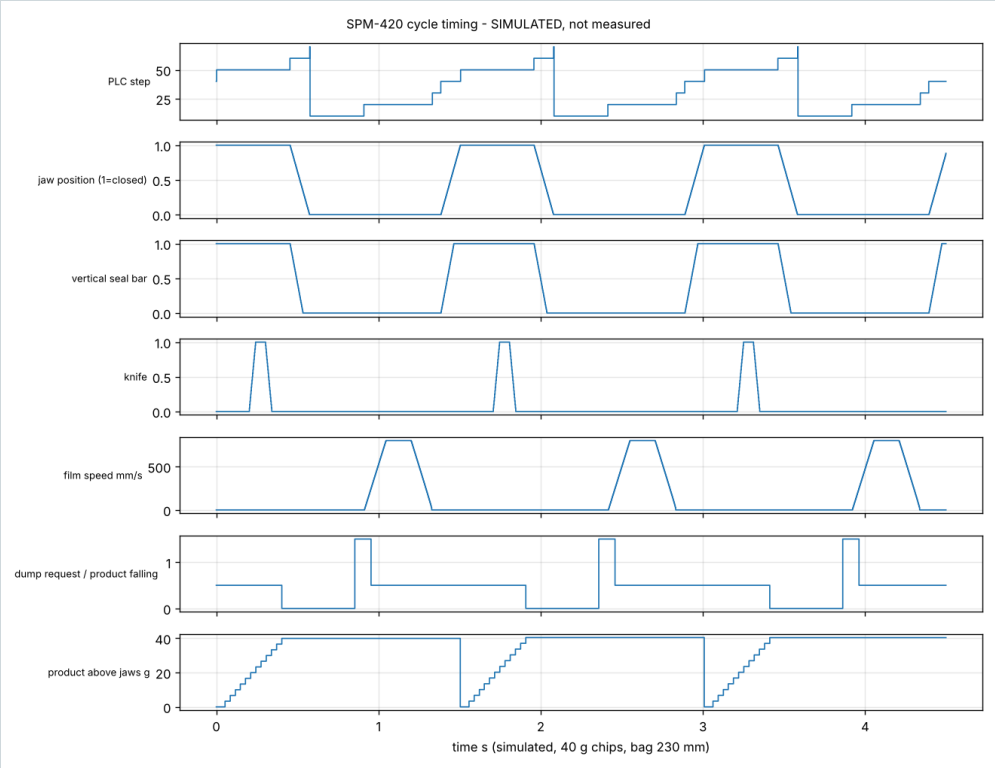
AREA	STATE	MEANING
Concept, layout, footprint	`Complete`	Derived from selected components; final heights wait for weigher and elevator supplier drawings.
Calculations	`Complete`	Cycle, drive, pneumatics, heating, electrical load, consumption. Based on stated assumptions.
3D model and drawings	`Layout level`	Correct envelopes and interfaces; detailed part drawings with tolerances still to be made.
Electrical design	`Design level`	Architecture, protection, safety circuits, I/O, cables; production schematics still to be drawn.
PLC program and HMI	`Simulated`	Logic tested in simulation; not yet compiled in Siemens TIA Portal.
Monitoring software	`Runs`	Tested with simulated machine data; connection to a real PLC not yet tested.
Performance figures	`Calculated / simulated`	No figure has been measured. Validation needs the built machine.
Safety	`Draft`	Concept and preliminary risk assessment; must be completed and validated on the built machine.

2. How the machine works

Hopper and tray	Z-bucket elevator	Multihead weigher	Forming and pull	Seal and cut	Take-off conveyor
Meters product onto the elevator. `Runs on weigher demand`	Lifts product about 3.5 m. `PLC, on weigher demand`	Picks the heads whose sum is closest to target and drops them. `Weigher's own controller`	Film is wrapped into a tube and pulled down one bag length. `PLC + servo`	Jaws seal the bag top and the next bag's bottom; a knife cuts between. `PLC + valves`	Carries the finished bag past a counting photo-eye. `PLC`

The bagger works in a fixed cycle. For the reference recipe (40 g chips, 230 mm bag, 40 bags/min) one cycle takes 1.5 s. The times below come from the simulation of the PLC program and are design values, not measurements.

STEP	PHASE	TIME, S	WHAT HAPPENS	CONTROLLED BY
10	Dump request	−0.056	Shortly before the film pull, the PLC tells the weigher to drop the next charge. The charge needs about 0.70 s to fall to the jaws.	PLC, weigher
20	Pull film	0.000–0.422	Pull belts draw 230 mm of film down over the forming tube.	Servo drive
30	Settle	0.422–0.474	Short pause so the film is still.	PLC timer
40	Close jaws	0.474–0.596	Horizontal jaws and the vertical seal bar close.	PLC, valves
50	Seal and cut	0.596–1.048	Two seals are made at once; the knife cuts between them at 0.798 s. The new charge lands on the closed jaws from 0.646 to 0.996 s.	PLC timer
60	Open jaws	1.048–1.170	Jaws, seal bar and knife retract; the finished bag is on the conveyor.	Spring-return valves
10	Wait	1.172–1.504	The PLC holds until the next cycle is due and the weigher is ready.	PLC pacing



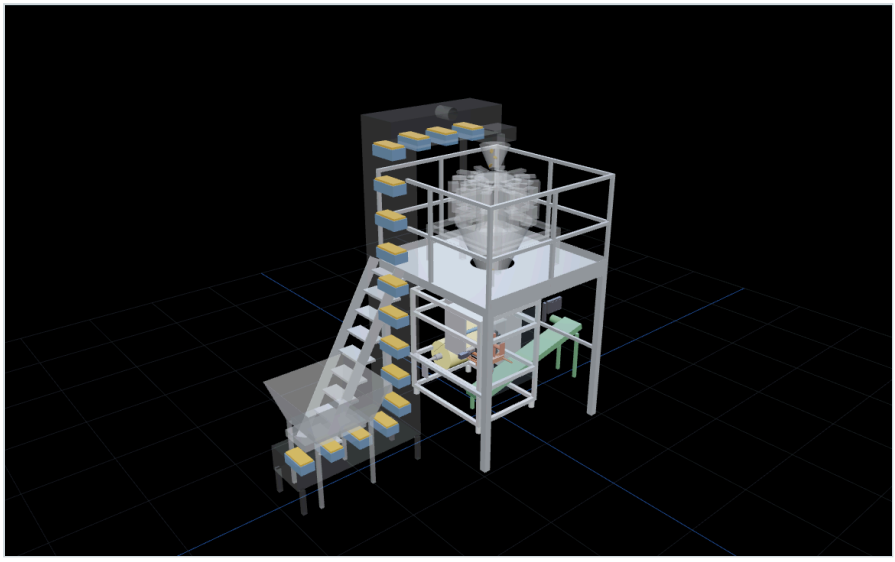
Simulated timing of three consecutive cycles: PLC step, jaw, seal bar, knife, film speed, dump request and product above the jaws.

Safety is independent of the software

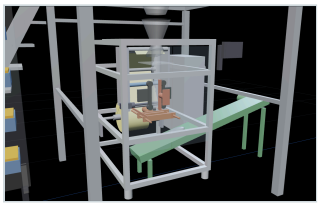
Emergency stop, guard-door interlocks, the servo's safe torque off and the heater over-temperature cut-out act directly on the valves, drive and contactors. The PLC only reports their state and cannot override them. There is no bypass function.

See it run: the 3D simulation

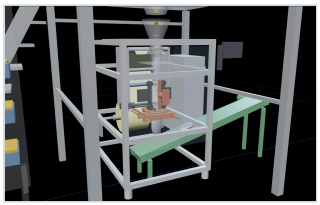
The line can be watched working on an interactive 3D model at **spm-420.vercel.app**. Press **Run** in the Simulation box. No login or software is needed.



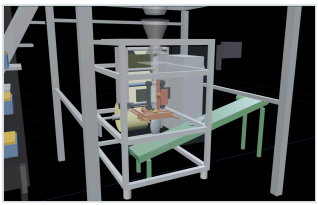
The whole line during the simulation. Buckets carry product up the elevator; the selected weigher heads light up blue; finished bags leave on the take-off conveyor. The elevator, weigher and guards are shown see-through while it runs.



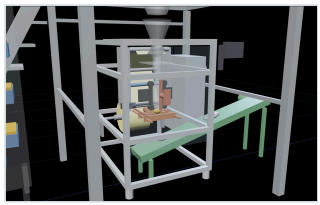
1. Pull film (0.15 s). Jaws open; the filled bag moves down below the jaws while the next charge falls.



2. Close jaws (0.50 s). The jaws and the vertical seal bar close on the film.



3. Seal and cut (0.87 s). Two seals are made, the knife cuts, and the bag is released.



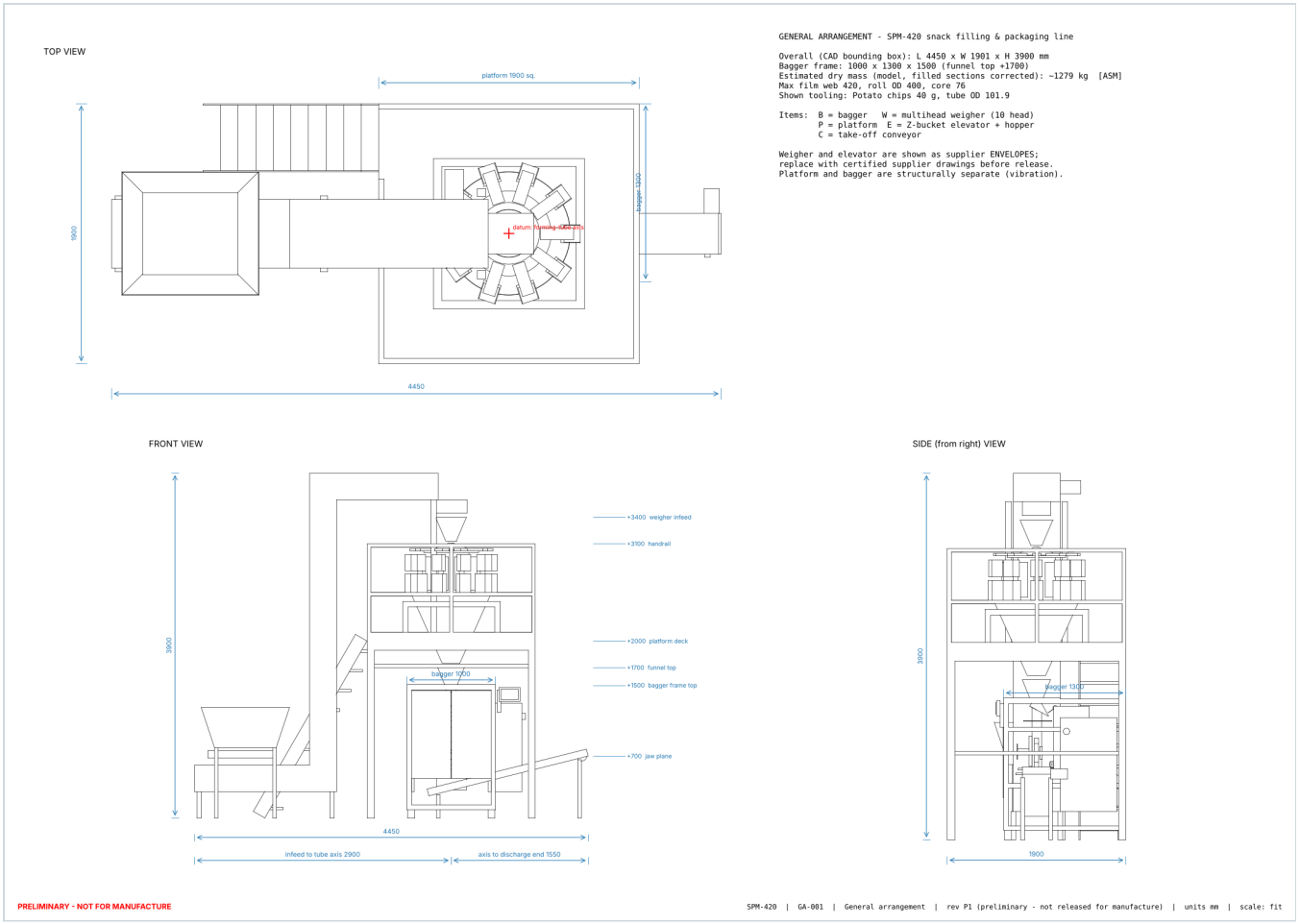
4. Discharge (1.30 s). Jaws open; the bag rides up the conveyor; the next dump is about to be requested.

ON THE PAGE	WHAT IT DOES
Run / Pause / Reset, speed	Plays the process at 0.25×, 0.5× or real time. A readout shows the PLC step, cycle time, jaw state, bags made and rate (39.9 bags/min).
View buttons	Iso, front, side, top, and a close-up of the bagger. Drag to rotate, scroll to zoom.
Explode slider, part list	Pulls the sub-assemblies apart; shows, hides or isolates each part; clicking a part shows its material, size and estimated mass.
Downloads	This overview, the component sheets, and the 3D model as STEP and GLB.

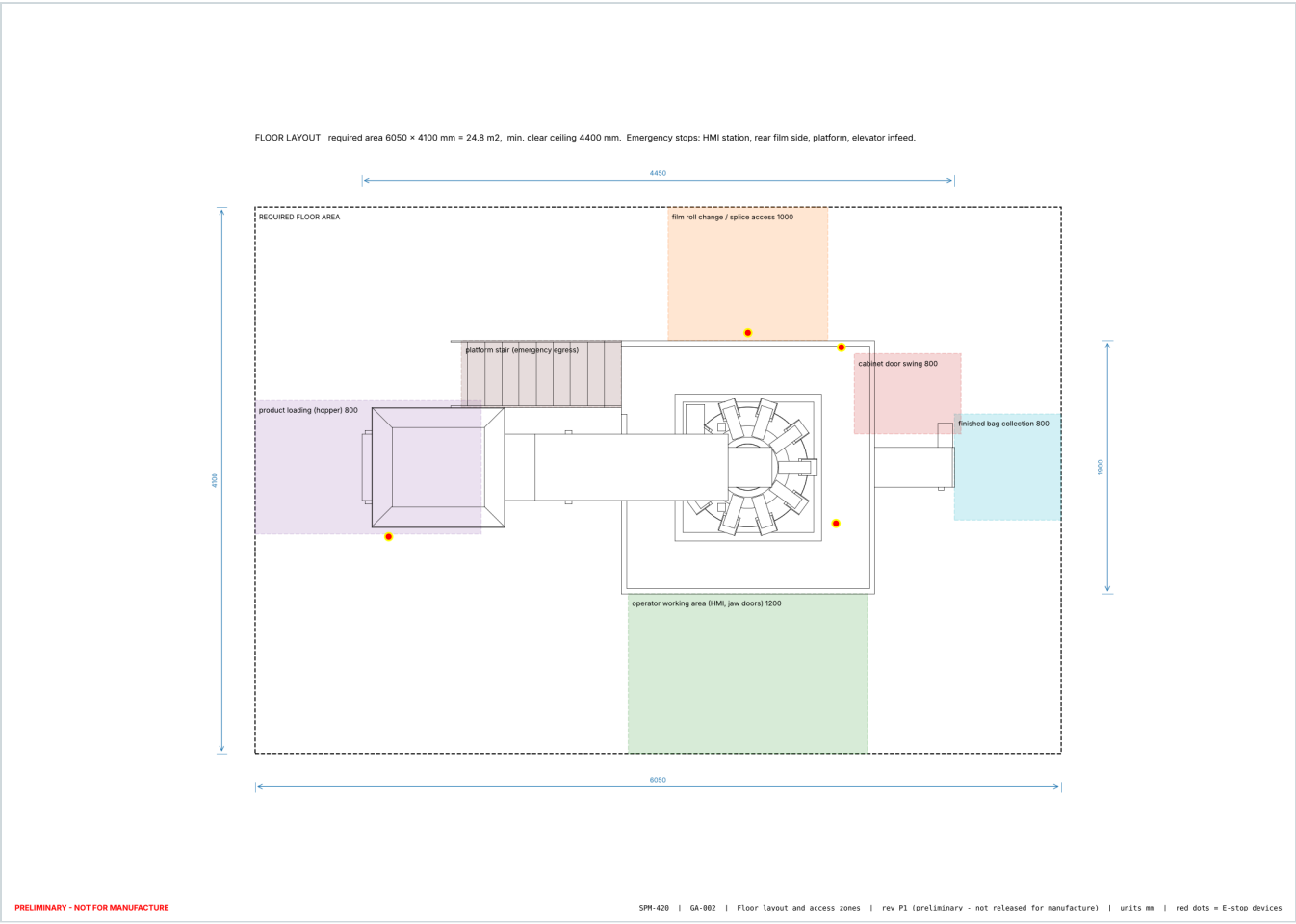
What the simulation is

The movements of the jaws, seal bar and film replay the simulation of the PLC program. The buckets, falling chips and bags are illustrative and follow the same timing. It shows how the design is intended to work; it is not a recording or a measurement of a machine.

3. Layout and dimensions



GA-001 General arrangement: top, front and side views with overall dimensions and key heights.



GA-002 Floor layout: required area, access zones and emergency-stop positions (red).

DIMENSION	VALUE, MM	SET BY
Line length	4450	Elevator infeed hopper to end of the take-off conveyor
Line width	1900	Weigher platform; the bagger and film roll fit inside
Line height	3900	Elevator head above the weigher
Bagger frame	1000 × 1300 × 1500	Film width, film path, jaw stroke, bag drop height
Floor area with clearances	6050 × 4100	1200 operator front, 1000 film change rear, 800 at each end
Clear ceiling height	4400	Room to lift weigher hoppers and elevator parts
Weigher platform deck	+2000	Follows the weigher chute; supplier dependent
Jaw plane	+700	Conveyor height plus the longest bag

All dimensions are preliminary. They become manufacturing dimensions only after the weigher and elevator suppliers' certified drawings are received and the first assembly is checked against the layout.

4. What the design package contains

Everything below exists as files in the project repository and can be handed over. Editable source formats are included so that every item can be changed and regenerated.

DELIVERABLE	CONTENTS	FORMATS	STATE
Reference analysis	Machine principle, the ten requested identifications, published comparison data, checklist for confirming the video	Document	Inferred
Engineering specification	Functional, performance, utility and design requirements with IDs	Document	Complete
Engineering calculations	Bag geometry, cycle, servo sizing, sealing forces, air, heating, electrical load and cables, consumption, footprint, structure	Document + calculation script	Complete
Mechanical design	Choice of dosing method, every bagger sub-assembly with sizes, materials and mounting, fragile-product measures	Document	Layout level
3D model	Complete assembly with 22 named parts, exploded view, one file per part	STEP, STL, PNG, PDF, parametric source	Layout level
Drawings	General arrangement, floor layout, 22 component sheets	PDF, DXF, PNG	Layout level
Electrical design	Architecture, power distribution, safety circuits, motor, heater and sensor wiring, cable schedule, terminal plan, panel layout	Documents, CSV	Design level
Bill of materials	77 lines with specification, supplier class, model number where verified, price estimate, three configurations	CSV + budget script	Estimates
PLC program	Siemens S7-1200 Structured Text: bag cycle, film pull, temperature control, modes, alarms; 65 I/O points, 32 alarms	TIA Portal import sources, CSV	Simulated
HMI	Specification of 14 screens, 57 tags with limits and user levels, clickable browser prototype	Document, CSV, HTML	Prototype
Monitoring software	Production statistics, batch records, alarm history, maintenance tracking; read-only towards the PLC	Python, React, PostgreSQL, Docker	Runs
Weigher algorithm	Combination-weighing routine for a self-built prototype weigher	C source + tests	Tested on PC
Simulation	PLC logic against a machine model, 12 scenarios including faults; weigher statistics	Python, results	Complete
3D website with simulation	Interactive model with a running process simulation, part data, drawings and downloads at spm-420.vercel.app	Website, GLB	Live
Manufacturing and assembly guide	Fabrication, tolerances, tools, assembly sequence, film threading, format change, wear parts	Documents	Complete

DELIVERABLE	CONTENTS	FORMATS	STATE
Safety documentation	Standards list, preliminary risk assessment, lockout and cleaning procedures	Documents	Draft
Testing and calibration	Test plan with acceptance criteria, commissioning checklist, calibration procedures	Documents	Complete
Budget	Three configurations in USD and IQD	Document	Estimates
Engineering report	Findings, assumptions register, limitations, next steps	Document	Complete

5. What has been verified, and how

CHECK	RESULT	WHAT IT DOES NOT PROVE
PLC logic against a machine model: normal running, warm-up, empty hopper, film out, wrong weight, jams, servo fault, over-temperature, sensor failure, power loss, emergency stop, air loss, door opening	12 scenarios, 51 checks, all pass	That the Siemens source compiles, or that real actuators and product behave like the model
Combination weighing at 40 g (20,000 simulated dumps)	Mean 40.5 g, standard deviation 0.7 g, 1.4 % give-away, no bag below the legal tolerance	Real weigher performance: depends on the product and the supplier's unit
Weigher algorithm against brute force (20,000 random cases)	Identical results	Behaviour on a microcontroller
Monitoring software: 8 automated tests plus a live browser run	Pass	Connection to a real PLC
HMI prototype: login, all screens, start, emergency stop, reset	Pass in a browser	The real Siemens HMI project
3D model against the calculated footprint	Bounding box equals 4450 × 1900 × 3900 mm	Detailed interferences and manufacturability

Defects found and fixed before any hardware existed

1. The machine could hang in the stopping state after an aborted cycle.
2. Alarms needed two presses of RESET to clear.
3. The recipe speed was not enforced.
4. The weigher was told to drop too late. The charge landed after the jaws had reopened, on a hot seal held only by the film. The drop is now requested shortly before the film pull, and in the simulation the whole charge lands on the closed jaws with about 50 ms margin on each side.
5. The PLC reported 51.3 bags/min while the machine ran at 40.

Finding these is the purpose of the simulation. It does not show that no other defects remain.

6. What the project must supply

A design package cannot replace physical equipment, licensed software, qualified people or measurements. These items have to come from the project.

Equipment to buy complete

ITEM	REASON
10-head multihead weigher	A precision instrument with load cells, its own controller and legal-metrology approval. Economy class about 9,000 USD; branded and approvable units cost much more.
Z-bucket elevator with infeed hopper	Standard product; buying is cheaper than building. About 2,800–4,500 USD.
Forming collar and tube, one set per bag width	Specialist part; its shape decides whether the film runs straight.
Air compressor with dryer and filters	Only if the factory has no clean, dry compressed air.

Components to buy

Siemens PLC with I/O modules, 7" touch panel, servo drive and motor, two frequency inverters and motors, safety relays, emergency-stop buttons, coded door switches, sensors, pneumatic cylinders and valves, cartridge heaters, thermocouples, stainless enclosure, cables and terminals. All are in the bill of materials. Three order numbers there are marked for confirmation by the distributor.

Fabrication

- Stainless-steel frame and weigher platform: TIG welding workshop.
- Sealing jaws: machined from tool steel, hardened, ground and nickel-plated.
- Pull-belt units, rollers, film unwind, sheet-metal parts: lathe, mill, laser cutting and bending.

Software licences and tools

- Siemens TIA Portal (STEP 7 Basic with WinCC Basic) and a PC to run it: needed to compile the PLC program and build the HMI.
- OPC UA runtime licence for the PLC, only if the monitoring PC is used.
- SISTEMA (free) for the safety performance-level calculation.

Qualified people

ROLE	TASK
Mechanical designer	Review the part drawings and sign them for manufacture.
Welder and machinist	Fabrication to the drawings and tolerances.
Electrician / panel builder	Build and test the cabinet and field wiring to IEC 60204-1.
PLC engineer	Compile, download and commission the program on the machine.
Weigher supplier technician	Install, calibrate and tune the weigher for each product.
Safety assessor	Complete the risk assessment and validate the safety functions on the built machine.

ROLE	TASK
Conformity body	Local approval (COSQC in Iraq), or CE assessment if the machine is exported.

Measurements that replace assumptions

MEASURE	WHY IT MATTERS
Bulk density of each product (1 L cylinder, five repeats)	Bag size
Film data sheet and seal trials on a test jaw	Seal temperature and time; whether the jaw cylinder is strong enough
High-speed video of the product falling (a 240 fps phone is enough)	Setting the weigher drop time
Film pull force (spring scale)	Confirms the servo selection
Supply voltage, earthing system, summer hall temperature	Electrical protection and cabinet cooling
Supplier quotations and certified drawings	Replace price estimates and preliminary dimensions

Site and consumables

Floor space of about 6 × 4 m with 4.4 m clear height; 400 V three-phase 16 A supply; compressed air at 0.6 MPa; film rolls and product for trials; certified test weights; spare knives, PTFE tape, belts and heaters.

7. Items that can still be added to the package

Several items listed above as needing a specialist tool can still be prepared, at least in part, as documents. Each would shorten the work of the people in section 6 but still needs their review.

ITEM	WHAT CAN BE PRODUCED	WHAT STILL NEEDS A PERSON OR TOOL
Part drawings for manufacture	Detailed drawings with dimensions, tolerances, materials and surface finish for the machined and sheet-metal parts (jaws, carriages, rollers, shafts, funnel, chutes), generated from the parametric model	Review and signature by a mechanical designer; confirmation against the purchased forming set
Electrical schematic sheets	Numbered schematic pages for power, safety, PLC I/O and motors as PDF and DXF, drawn from the existing design	Check against the actual devices' manuals by an electrician; redraw in an ECAD tool if the panel builder requires one
TIA Portal import files	PLC tag table, HMI tag list and alarm texts as Excel files for TIA Portal's import function	The import itself and the HMI screen building in TIA Portal; format checked against the TIA version used
Independent syntax check of the PLC logic	A port of the program to standard IEC 61131-3 Structured Text, compiled with an open-source compiler if one can be installed	Final compilation in TIA Portal, which only a Siemens licence can do
Safety calculation input	Prepared SISTEMA input: safety functions, structure, component list with the reliability values from manufacturer data sheets	Running SISTEMA and validation on the machine by a competent person
Requests for quotation	Ready-to-send technical requests for the weigher, elevator, forming sets, fabrication and panel build, with the interface data each supplier needs	Sending them and negotiating
Video and photo analysis	Frame-by-frame analysis of videos you send: the reference clip, product fall timing, seal trials	Recording the videos
Operator documentation in Arabic	Operator instructions, HMI texts and safety labels in Arabic	Review by a native-speaking operator
Updated design from real data	Recalculation of drawings, code and budget once measurements and quotations arrive	Providing the data

> **What no document can replace**
>
> Physical parts and equipment, licensed Siemens software, legal approvals and certificates, the safety validation of the built machine, and measured performance figures.

8. Estimated budget

Prices are estimates made without supplier quotations, except two items whose range comes from public supplier listings. Exchange rate 1 USD = 1320 IQD (Central Bank of Iraq cash selling rate, November 2026). Freight, customs and local margins are covered only by the contingency line.

CONFIGURATION	USD	IQD	ACCURACY
1. Educational prototype	13,993	18,471,024	±40 %
2. Semi-industrial machine	**55,919**	**73,812,816**	±30 %
3. Industrial machine	135,083	178,309,296	±35 %

	1. EDUCATIONAL PROTOTYPE	2. SEMI-INDUSTRIAL	3. INDUSTRIAL
Purpose	Teach and prove the sequence	Small factory production	Continuous production with quality records
Frame	Aluminium profile, bench size	Welded stainless steel	Welded stainless, hygienic detailing
Weighing	Single self-built weigh head	Economy 10-head weigher	Branded 10/14-head, approvable for trade
Control	S7-1200 and 4" panel	S7-1200, 7" panel, safety relays	Fail-safe S7-1200, 7" panel, monitoring PC
Quality devices	None	None (manual sampling)	Checkweigher, metal detector, date coder, nitrogen flush
Speed (calculated)	8–15 bags/min	30–45 bags/min	40–60 bags/min
Selling food	No	After safety validation and hygiene review	Yes, after conformity assessment

Running cost (calculated, semi-industrial at 40 bags/min): energy and film ≈ 25.8 USD per hour, about 12.7 USD per 1000 bags. Film is about 98 % of this figure, so bag length matters far more than energy use. Labour, maintenance and depreciation are not included.

9. Open issues and assumptions

ISSUE	DETAIL	HOW IT IS CLOSED
Jaw sealing force	The selected cylinder gives 1683 N against an assumed need of 2304 N. Three remedies are prepared: narrower seal lands, a larger cylinder, or a toggle linkage. The frame is sized for the largest option.	Seal trials on the real film
Weigher drop timing	The drop is set 55 ms before the film pull. In the model this works between about 10 and 100 ms.	High-speed video per product
Weight errors	The bagger cannot detect a wrong bag weight; this depends on the weigher and, for products sold by weight, on a checkweigher.	Decision on the quality configuration
PLC program	Not yet compiled in TIA Portal; first import may show syntax errors.	Compile and bench test
Speed limit	Pneumatic jaws limit the reference bag to about 49 bags/min. Faster speeds need servo-driven jaws, a different mechanism.	Decision if a higher speed is required
Cabinet temperature	About 45 °C inside at 35 °C hall temperature. A filter fan or cooler is specified for hot halls.	Site survey
Reference machine	The configuration is inferred, not observed from the video.	Confirm against the clip
Product and site data	Bulk densities, film properties, fall times, supply and earthing are assumed.	Measurements in section 6

10. Recommended next steps

1. **Confirm the scope:** products, bag sizes, speed, quality devices and configuration (section 8).
2. **Request quotations** for the weigher and elevator, including certified drawings.
3. **Build a jaw test rig and run seal trials** on the chosen film. This settles the jaw-force question cheaply, before the frame is built.
4. **Detail design:** part drawings and electrical schematics; structural check of the weigher platform.
5. **Bench test** the PLC, servo and touch panel together before installation.
6. **Build and commission** following the commissioning checklist.
7. **Validate safety**, then run the performance tests and replace every calculated figure with a measured one.

> **Decisions needed from the client**
>
> Product list and bag sizes · target speed · semi-industrial or industrial configuration · whether a checkweigher and metal detector are required · site data (supply, earthing, hall temperature, ceiling height).